Photovoltaics, Humans and the Biosphere: A transdisciplinary approach fostering Alpine resilience (BioPV)

Final Report, February 2026

Scherhaufer, Patrick; Baumgartinger, Mathias; Bittner, Karl; Dworzak, Veronika; Hainz-Renetzeder, Christa; Hampl Nina; Klic, Rafaela; Mikovits, Christian; Neumayer, Johann; Pachinger, Bärbel; Plöchl, Jana; Pointner, Daniel; Schauppenlehner, Thomas; Scheibenreif, Silvia; Schiesser, Beatrix; Sposato, Robert; Wiesenhofer, Florian

1. Project overview

- Project Title and Acronym
- Project Duration
- (Budget)
- PIs and Team Members – Stage of career, Affiliations, and roles in the project
- Institutions
- Involved Stakeholder

2. Summary / Conclusio

Executive Summary – German and English-Policy letter if applicable – Conclusion

3. Project objectives and goals

- Initial Objectives – Clearly state the original goals and aims of the project
- potential deviations from planned project
- Key Research Questions
- Interdisciplinary and Transdisciplinary Approach
- Connex to international Programs (UNESCO, UN, EU, ect)

4. Scientific results and findings

4.1. WP2

Material and methods

Study area

The analysis was conducted across five Austrian biosphere parks representing diverse Alpine, pre-Alpine and floodplain alluvial lowland conditions: *Großes Walsertal* (192 km²), *Lungau* (1,009 km²), *Nockberge* (485 km²), *Unteres Murtal* (132 km²), and Wienerwald (1,056 km²). These protected areas combine conservation goals with sustainable regional development, making them ideal case studies for integrating renewable energy deployment with ecological and social considerations. Additionally, calculations based on the methodologies and criteria developed and explored during this work package were performed for all of Austria.

Siting criteria and literature review

As a first step we performed a literature review which investigates the current state of knowledge on siting criteria for ground-mounted photovoltaic systems (GMPV). We examined a comprehensive range of factors influencing optimal PV site selection, including environmental, economic, technical, but also socio-political considerations. The review analyses existing methodologies, tools, and models employed in site assessment and prioritization, with a particular focus on the integration of geographic information systems (GIS) data. By identifying knowledge gaps this study aims to contribute to the development of more effective and sustainable PV deployment strategies, especially in areas with small landscape structures and sensitive areas, such as biosphere reserves.

We identified over 6,000 scientific publications and selected more than 400 (n = 448) for detailed analysis and processing of siting criteria. Based on the literature research we created a comprehensive overview of parameters when determining the location for ground-mounted PV systems, categorized into five main areas: solar radiation; land & topography; economics; infrastructure; and natural hazards.

Stakeholder workshop

In addition to the criteria derived from the existing literature, the participants of the stakeholder workshops were also asked to provide input, in particular regarding the weighting of criteria. In the BioPV workshops, selected criteria were qualitatively assessed, including panel density (6–30 m²/kWp), plant size (0.5 ha to >5 ha), and distance to settlements. The three most important results were (i) a preference for plants in proximity to residential areas, (ii) a tendency towards fewer large plants rather than many small ones, and (iii) a medium-to-loose panel density to enable biodiversity measures, and these preferences were considered in the techno-economic analysis via connection costs and scale effects.

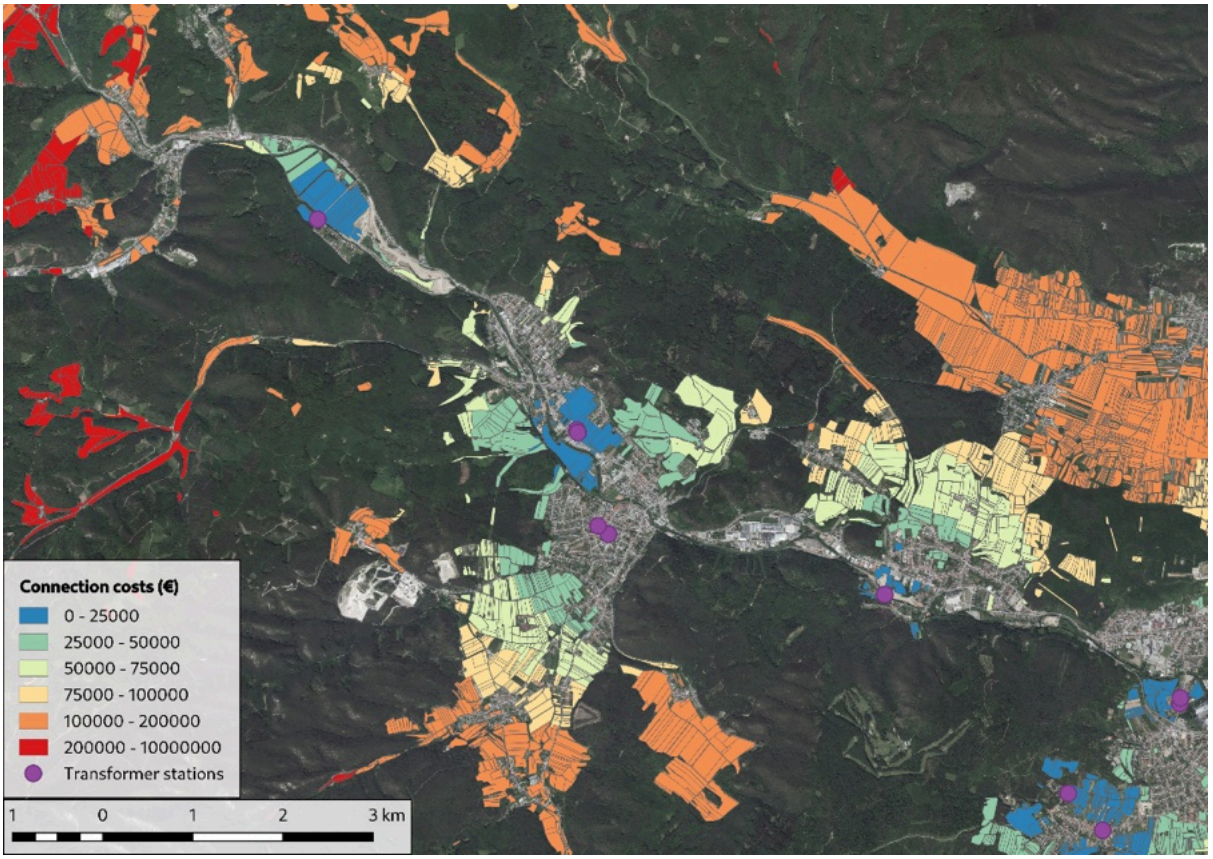
Simulation Framework

The framework integrates geospatial tools and algorithms to process and analyze several different datasets, e.g. topography, land use, solar irradiation and electrical infrastructure such as transformer stations. As we are using open data sources to promote transparency and replicability, open data for especially the electrical infrastructure relies mainly on community driven data from open streetmap (OSM) and is far from complete. We operationalized the suitability and techno-economic feasibility through a differentiated assessment system that links technical and economic factors and reports results under three variants (Maximum, Medium, Minimum).

hier Tabelle1 aus dem Factsheet inklusive Datenquellen

The modular framework ingests gridded and vector geodata and harmonizes them into a common CRS and extent. It derives criterion layers and applies the scenario-specific thresholds and cost uplifts from table xx. We finally compute parcel-level outputs, eligible and not eligible, simulated annual generation estimates, cost components (CAPEX, OPEX), and LCOe.

Figure XX illustrates connection costs in a mountainous region, showing how costs increase rapidly with distance from suitable transformer stations. This demonstrates that systems located far from settlements are typically economically infeasible.



Results

The simulations indicate substantial GMPV potential across the five biosphere study regions across all three scenario variants. In the Maximum scenario, permissive thresholds identify suitable agricultural land (alpine pastures, grassland, and arable land) up to more than one hundred square kilometers per park, translating into multi-TWh annual generation. As constraints and cost caps tighten in the Medium and Minimum variants, the technically viable area contracts; however, the remaining sites offer improved economic profiles with lower LCOe. Ultimately, the conservative scenario still yields a multi-TWh annual potential, suggesting that biosphere reserves can contribute significantly to energy targets without the need for marginally economic sites.

Spatially, the results highlight marked differences between biosphere parks. Regions such as Nockberge and Unteres Murtal combine favourable solar resource, relatively low forest cover and short average distances to substations, which leads to both comparatively high area potentials and low LCOE across all scenarios. By contrast, more mountainous or densely forested parks show a stronger sensitivity to the scenario assumptions: increasing slope-related cost uplifts and compactness penalties removes a larger proportion of parcels from the feasible set. These patterns underline the importance of explicitly representing terrain, land-use structure and grid topology in PV siting, as purely radiation-based screening would overestimate the economically usable potential.

Hier die Grafik und Tabelle2 aus dem Factsheet 1

Additionally, the analysis of rooftop and parking-lot PV indicates that already sealed surfaces can provide a non-negligible supplementary contribution to regional PV generation, particularly in more urbanized biosphere parks. While the absolute potentials on buildings and car parks are small compared to ground-mounted scenarios, they are associated with minimal additional land-use conflicts and can therefore play a strategic role in integrated deployment strategies. In aggregate, the combination of high-quality ground-mounted sites and targeted use of sealed areas offers a robust pathway to upscale PV in biosphere reserves while respecting both economic constraints and competing land-use objectives.

Hier noch die Ergebnistabelle 3 aus dem Factsheet 1

4.2. WP3 - Biodiversity promotion

- Main Research Outcomes – most significant discoveries and results

Literature review on biodiversity impacts of ground-mounted PV

Scientific literature on the impact of ground-mounted PV on biodiversity is scattered and scarce (Gómez-Catasús et al. 2024, Lafitte et al. 2022). In BioPV, we added to these reviews and focused on the Central European conditions to specify the knowledge gaps. Additionally we evaluated on how the possible biodiversity impacts of ground-mounted PV systems are addressed in spatial planning and spatial siting criteria in Austria. Direct impacts on biodiversity still show large variation and ambiguous results (Table 1), thus planning and the formulation of mitigation criteria also varies a lot.

Table 1: Number of articles dealing with indirect and direct impacts on biodiversity; articles include statistical as well as conceptual and review papers, number of statistical papers in brackets. Biodiversity level studied include different biodiversity indicators as well as spatial models. (Hainz-Renetzeder et al., 2025a)

Impacts	Number of articles	Biodiversity level studied	Clarity of results
Alteration of environmental conditions	24 (16)	Site condition properties	Clear results of altered conditions
population composition & diversity	31 (25)	plants, arthropods, bats, birds	large variation in results
habitat loss, degradation and fragmentation	24 (15)	protected areas, wilderness areas, land cover, modelled habitats	ambiguous results
impact on movement and behaviour (resident & migratory species)	7 (4)	pollinators, butterflies, bats	ambiguous results
mortality & collisions	17 (7)	birds, bats, aquatic insects	large variation in results
exotic plant invasions	5 (1)	non-native plants	higher cover of neophytes

Spatial siting criteria for ground-mounted PV in Central Europe are guided by the mitigation hierarchy (Kiesecker et al., 2010): first, avoid high-conservation-value areas, then reduce impacts where development proceeds. Guidelines consistently exclude legally protected areas, especially those with high IUCN status; but treatment of valuable habitats outside protected areas is inconsistent across the spatial planning guidelines (Hainz-Renetzeder et al., 2025a). Some do not address this issue, while others vary between exclusion and conflict (on-site assessment) zones. Some Austrian planning frameworks label “suitability zones”, but inclusion of nature conservation criteria differs greatly among provinces and guidelines. The different regulations within Austria lead to uncoordinated and non-transparent integration of nature conservation into siting of ground-mounted PV systems, underscoring the need for a harmonized, evidence-based approach and comprehensive ecological data.
Overall, strategic, standardized conservation planning is required to balance energy expansion with habitat protection in siting decisions.
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Integration of ecological parameter into techno-economic assessment
In order to overcome this drawback, we used the frame of “High-Nature-Value farmland”, a concept emerging in the 1990’s (Baldock et al., 1993), which acknowledges that many species are dependent on traditionally extensively used cultural landscapes and related semi-natural elements (Parrachini et al., 2008). Based on the IACS dataset used in WP2 for the techno-economic assessment, we identified the IACS agricultural uses most likely being available for HNV based on categories used by Bartel et al. (2011). Thus, we identified areas where high PV potential conflicts with high HNV values (conflict areas) or not (priority areas).

The results show that there are large differences among the biosphere reserves, with “Unteres Murtal” having large areas with high PV potential and low HNV values available (Fig. XX). Also, the Lower Austrian part of the Wienerwald biosphere reserve shows a high

amount of priority areas. The share of highly valuable farmland with high PV potential is rather small and should therefore be excluded from any construction activities.

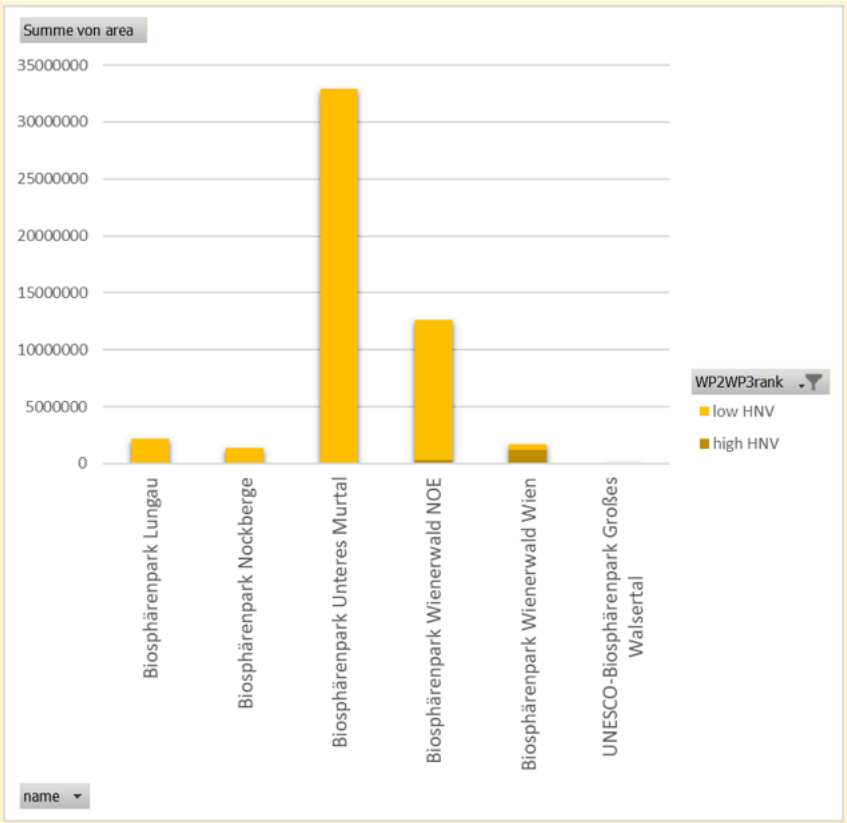

Figure XX: Total area in m2, where high PV potential is overlaid with HNV-values, shown for each biosphere reserve. Low HNV values indicate areas where the installation of ground-mounted PV systems should be prioritised in a manner compatible with biodiversity (priority areas). On the other hand, construction of PV-systems should be avoided in areas with high HNV (conflict areas).

Field work and sampling of organism groups

This superficial level helps to formulate general recommendations, but lacks detailed insight into the differences among the many varieties of designing, implementing and managing ground-mounted PV installations. The scientific evidence on the impacts of certain organism groups is still scarce. In order to gain more insight into these details, the indicator groups of wild bees, grasshoppers, and butterflies were studied in three case study biosphere reserves using field surveys in 2023 and 2024. The nine selected ground-mounted PV systems varied in size (400 m² to 30,000 m²) and land use history (e.g., arable land, grassland). Nine control areas with similar surroundings and land use served as comparisons. A total of 122 wild bee species (17% of the 700 species found in Austria), 46 butterfly species (36% of Austria's 208 butterflies) and 39 grasshopper species (19% of Austria's 127 species) were observed during the field surveys (Hainz-Renetzeder et al., 2025b).

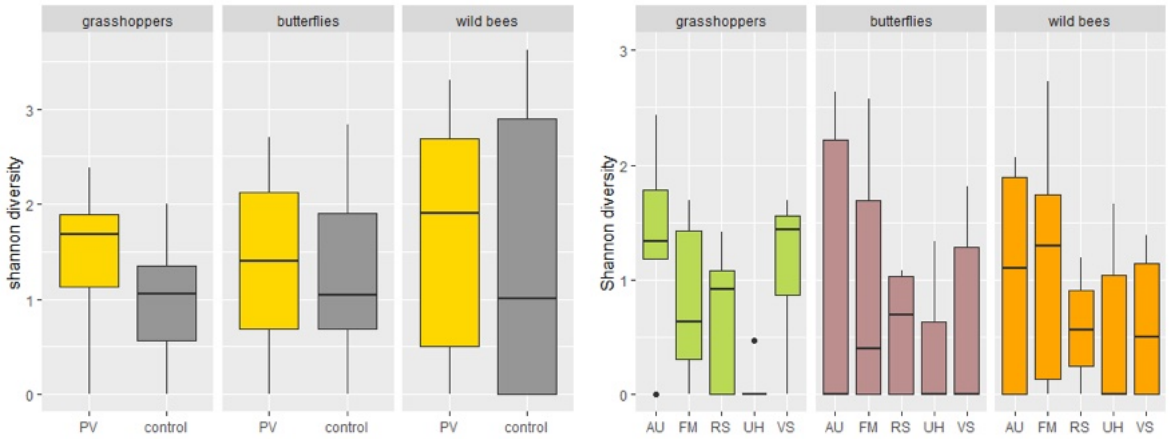

Figure XX: left) Shannon diversity index of the three indicator groups in the ground-mounted PV systems (PV) and reference areas (RE); right) Shannon diversity index of the three indicator groups in the different transects of the ground-mounted PV systems; AU: outside of the panel areas, FM: special structures for promoting biodiversity, RS: backside of the panels, UH: below the panels, VS: front side of the panels

The Shannon diversity index of all three organism groups showed a large variation, both inside the ground-mounted PV systems and the reference areas (Figure XX left). This can be explained by the varying design and management of the sites. Within the ground-mounted PV systems, it was found that the transects outside the rows of panels and the special structures for promoting biodiversity are particularly species-rich, while the areas below the panels tend to be species-poor. The front edge of the panels is rich in flowers and flower visitors in dry locations, but in moist locations, grasses dominate due to high nutrient input and there are fewer flowers than the back of the panels.

Based on these results and supported by literature, it is clear that ground-mounted PV systems can serve as potential habitats for various species if they are designed and maintained appropriately (BioPV webinar, Hainz-Renetzeder et al., 2025b). The results underscore the need to integrate ecological measures into the planning and maintenance of such facilities in order to promote biodiversity while ensuring energy production. In addition to careful site selection, the following recommendations can be formulated:

- Preservation and promotion of existing biotope structures
- Leaving areas unmaintained
- Creation of scrubby areas
- Creation of open ground areas
- Preserve and promote structural diversity: e.g., with deadwood, hedges, and stone piles
- Creation of perches and nesting habitats
- Creation of diverse flowering areas through site-adapted, alternating mowing
- Creation of diverse flowering areas through sowing with species-rich, autochthonous seeds
- No construction work during breeding and spawning seasons
- Maximum 40-50% area coverage by the panels
- Ideally no fencing, if unavoidable - minimum distance to the ground 15-20 cm - avoidance of barbed wire

- If greening of the fence elements is planned: Privacy hedge with a width of approx. 5 m, planted with plants native to the area that are adapted to the location. Rows of trees, hedges or ditches are preferable alternatives to fences.
- Migration corridors: Crossing aids for animals or embedding of corridor areas in large-scale facilities
- Minimization of soil compaction and displacement

Monitoring recommendations need to be formulated for either the biosphere reserve management or associated partners to observe developments of biodiversity establishment in the PV installations which have to be targeted to the specific site conditions. These specific conservation targets from the flora and fauna of the respective ground-mounted PV site need to be defined for this purpose. On the one hand, these may be protected species (Red Lists, FFH). On the other hand, indicator species are identified to assess the effectiveness of implemented biodiversity-enhancing measures, as demonstrated in the bioPV project. For example, the presence of the Blue-winged grasshopper (*Oedipoda caerulescens*) serves as an indicator of the effectiveness of open soil patches. The Large Conehead (*Ruspolia nitidula*) indicates the success of shrubby or partially overgrown areas. Small Carpenter bees (*Ceratina* spp.) are used as indicators for the availability of standing dead stems, while Blues-species (Lycaenidae) reflect the quality of species-rich sown areas. In this way, the maintenance and development of floral diversity and structural heterogeneity can be effectively monitored and evaluated.

- Evaluation of results in relation to international research and recommendations for implementing ground-mounted PV in Austrian Biosphere Reserves

The sustainable development of climate change mitigation measures is a good example where biosphere reserves act on the forefront for establishing best-practice examples. The biodiversity-compatible planning, design and management of ground-mounted PV systems is also a topic internationally, e.g. biosphere reserve Rhön (Germany)[1]. In scientific literature, Nordberg & Schwarzkopf (2023) as well as Tölgyesi et al. (2023) describe a possible design and associated research gaps for the concept of an “ecovoltaic” (Tölgyesi et al., 2023) or “conservoltaic” (Nordberg & Schwarzkopf, 2023) park as a basis for establishing win-win-situation for biodiversity and energy generation. Still a point of criticism is the largely hypothetical approach, as there are major gaps regarding an effective implementation (Tölgyesi et al., 2023): What is the optimal shape and size of a solar park? What is the perfect panel spacing, panel width, panel height? For which organism groups do solar parks act as a barrier? Which seed mixtures and which management fit the local conditions and can effectively generate multiple ecosystem services for the surrounding landscapes? Which factors can be specified generally, and which differ from site to site?

The German Federal Agency for Nature Conservation (2024) stresses the fact that the site conditions, the immediate surroundings and the species found there also play a major role in the development of biodiversity in the ground-mounted PV systems. Not every species can be settled everywhere. From a nature conservation perspective, the design and maintenance of ground-mounted PV systems should therefore be based not only on

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1 https://www.biosphaerenreservat-rhoen.de/mensch/energie-und-klimaschutz/dialog-veranstaltung-freiflaechen-pv-zusammenfassung

ecological criteria but also on target biotopes and species. Target species are species of high conservation importance that need to be given priority protection and promotion in a particular region, for example, the skylark, whinchat, brown hare, or wild bees.

Basing the implementation of ground-mounted PV systems on these concepts, biosphere reserve could establish best-practice examples for targeted win-win-situations for sustainable climate change mitigation together with solid nature conservation activities.

- Publications – list of all published or submitted paper, date, journals, DOI

Hainz-Renetzeder, C., Schauppenlehner, T., Scherhaufer, P., & Pachinger, B. (2025a). Navigating the Nexus between Renewable Energy and Biodiversity: Impacts and Mitigation Strategies for Ground-Mounted Photovoltaic Systems. Landscape Online, 1141–1141. https://doi.org/10.3097/LO.2025.1141

Hainz-Renetzeder, Christa, Pospisil, Katharina, Neumayer, Hans, Pachinger, Bärbel (2025b): Ergebnisse der Freilanderhebung und Empfehlungen zur Förderung der Biodiversität, BioPV. Factsheet #2, Universität für Bodenkultur Wien, https://doi.org/10.5281/zenodo.17881407

in preparation:
Karner, K., Hainz-Renetzeder, C., et al. (in prep.): Experts' and stakeholders' perceptions on biodiversity impacts of ground-mounted photovoltaic systems and related conservation measures for Austria (Working title)

Hainz-Renetzeder, C., Mikovits, C., Pachinger B., et al. (in prep.): Conflicts between nature conservation and photovoltaic expansion on high-value farmland: An Austrian case study of affected habitats (Working title)

Hainz-Renetzeder, C., Pachinger, B., et al. (in prep): The importance of design and management of ground-mounted photovoltaics for arthropods (Working title)

- Data and Methods – Gathered data and used methods

Literature review
A systematic literature search was carried out via Scopus for the period 2012 to 2023 using a more general and a more specific search string. For the resulting articles, a word count of the key words were carried out and only those further processed of three or more hits. After screening the abstracts, 87 articles remained. This bibliography was was imported into research rabbit (https://researchrabbitapp.com/; Research Rabbit, s.a.), an AI-based tool that finds, organizes, and understands literature by visualizing a network of scientific articles and their connections, to ensure that all relevant literature has been found. This double-check resulted in adding another seven articles into the database with finally 93 articles. For more details on the literature review, see Hainz-Renetzeder et al. (2025a)

Integration of ecological parameter into techno-economic assessment
Based on the IACS dataset used in WP2 for the techno-economic assessment, we identified the IACS agricultural uses most likely being available for HNV based categories used by Bartel et al. (2011) and classified them into three categories. IACS categories classified as potentially highly valuable include mainly extensively used grassland, landscape structure elements such as hedges, ponds, or stone walls and fallow land. Valuable categories include mainly different types of grassland uses of medium intensity. All other categories include intensive agricultural activities (different types of arable land like crop, vegetable and energy biomass production). By overlaying both layers (i.e. PV potential and HNV farming), we could identify conflict potential in the biosphere reserves. Spatial analysis was done in QGIS 3.34 (QGIS.org 2023)

Field work and sampling of organism groups
We selected three existing ground-mounted PV systems for each biosphere reserve and reference areas for each of the nine ground-mounted PV plants that most closely corresponded to the landscape context and the previous use of the ground-mounted PV plants. Using the so-called transect method, different areas within the ground-mounted PV system (front edge, rear and below the panels, as well as the outer area and any special structures such as dead wood or hedges) and the reference areas were sampled for the three indicator groups (wild bees, grasshoppers, butterflies) in five survey rounds (summers 2023, 2024) and the organisms were recorded in terms of species and number. Statistical analysis was done in R 4.1.3 (R Core Development Team 2021; RStudio Team 2015).

Data for integration into the techno-economic modelling and the BioPVLabs
A list of GIS-Layers on ecological value was prepared as input into WP2 and the labs of WP5 in QGIS 3.34 (see Table XX). Some layers were defined as exclusion zones for ground-mounted PV systems, others were classified into their ecological value. For the Biosphere reserves Wiener Wald, Natura-2000-habitats were available, for the Lungau, a biotope mapping and the Red List of biotopes in Austria could be used for ecological classification.

Table XX: datasets used for classifying ecological value as suitable input into the techno-economic assessment and the BioPV labs.

dataset	data source	ecological classification
Naturschutzgebiete Ö 2024	UBA; Zitierung: Zusammenstellung durch Umweltbundesamt basierend auf INSPIRE-Services der Länder: © Land Burgenland, Land Kärnten, Land Niederösterreich, Land Oberösterreich, Land Salzburg, Land Steiermark, Land Tirol, Land Vorarlberg, Stadt Wien	exclusion
Lebensraumkorridore	UBA	exclusion

Feuchtgebietsinventar	UBA	exclusion
Auinventar	UBA	exclusion
Trockenrasenkatalog tw Revision	UBA	exclusion
Trockenrasenkatalog 1986	UBA	exclusion
INVEKOS 2023	BML	classification of ecological value
BPWW Offenlandkartierung	BPWW Management	classification of ecological value
Biotopkartierung Salzburg	Land Salzburg	classification of ecological value

References:

Baldock, D., Beaufoy, G., Bennett, G., & Clark, J. (1993). Nature conservation and new directions in the EC Common Agricultural Policy: the potential role of EC policies in maintaining farming and management systems of high nature value in the Community IEEEP London

Bartel, A., Süßenbacher, E., & Sedy, K. (2011). Weiterentwicklung des Agrar-Umweltindikators “High Nature Value Farmland” für Österreich. Umweltbundesamt, REP-0348.

German Federal Agency for Nature Conservation (2024): Solarparks und Biodiversität - Hintergrund; https://www.bfn.de/haeufig-gefragt-solarparks-und-biodiversitaet

Gómez-Catasús, J., Morales, M. B., Giralt, D., del Portillo, D. G., Manzano-Rubio, R., Solé-Bujalance, L., Sardà-Palomera, F., Traba, J., & Bota, G. (2024). Solar photovoltaic energy development and biodiversity conservation: Current knowledge and research gaps. Conservation Letters. https://doi.org/10.1111/conl.13025

Kiesecker, J. M., Copeland, H., Pocewicz, A., & McKenney, B. (2010). Development by design: Blending landscape-level planning with the mitigation hierarchy. Frontiers in Ecology and the Environment, 8(5), 261–266. https://doi.org/10.1890/090005

Lafitte, A., Sordello, R., de Crespin de Billy, V., Froidevaux, J., Gourdain, P., Kerbiriou, C., Langridge, J., Marx, G., Schatz, B., Thierry, C., & Reyjol, Y. (2022). What evidence exists regarding the effects of photovoltaic panels on biodiversity? A critical systematic map protocol. Environmental Evidence, 11(1), 36. https://doi.org/10.1186/s13750-022-00291-x

Nordberg, E. J., & Schwarzkopf, L. (2023). Developing conservoltaic systems to support biodiversity on solar farms. Austral Ecology, 48(3), 643–649. https://doi.org/10.1111/aec.13289

Paracchini, M. L., Petersen, J. E., Hoogeveen, Y., Bamps, C., Burfield, I., & van Sway, C. (2008). High Nature Value Farmland in Europe-An estimate of the distribution patterns on the basis of land cover and biodiversity data. https://data.europa.eu/doi/10.2788/8891

QGIS.org (2023). QGIS Geographic Information System. Open Source Geospatial Foundation Project. http://qgis.org.

R Core Development Team (2021) A language and environment for statistical computing, Vienna, Austria. https://www.R-project.org

RStudio Team (2015) RStudio: Integrated Development for R

Tölgyesi, C., Bátori, Z., Pascarella, J., Erdős, L., Török, P., Batáry, P., Birkhofer, K., Scherer, L., Michalko, R., Košulič, O., Zaller, J. G., & Gallé, R. (2023). Ecovoltaics: Framework and future research directions to reconcile land-based solar power development with ecosystem conservation. Biological Conservation, 285, 110242. https://doi.org/10.1016/j.biocon.2023.110242

4.3. WP4

- Main Research Outcomes – most significant discoveries and results
- Discussion – Evaluation of results in relation to international research
- Publications – list of all published or submitted paper, date, journals, DOI
- Data and Methods – Gathered data and used methods

This work package investigated how residents in and around Austrian biosphere reserves perceive utility-scale photovoltaic (USPV) projects when biodiversity, land use, and energy justice considerations are explicitly taken into account. Using a large online survey with an embedded choice-based conjoint (CBC) experiment, the study identifies land use type and biodiversity measures as key drivers of public support for USPV, and shows that social norms, activism intentions, and perceived impacts are central predictors of support for such projects. Walking interviews with local residents and stakeholders shed further light on the complex interplay of factors contributing to acceptance of ground-mounted photovoltaic installations on the individual level, revealing aspects of spatial and distributive justice as key factors contributing to local support, next to transparent decision-making and planning processes that take the local context into account.

Main research outcomes

The study shows that the way land is used for USPV is the single most important design factor for public support, followed by biodiversity measures and visual exposure of the infrastructure. Among the conjoint attributes, land use type had the highest relative

importance (35.86%), clearly outweighing biodiversity measures (21.65%), visibility (19.42%), procedural (12.57%) and distributional justice mechanisms (10.51%). Residents strongly preferred USPV projects on sealed or pre-impacted areas and near existing infrastructures (utility of 73.33), while projects on ecologically valuable land were strongly disfavoured (utility of –58.42), and installations on agriculturally used arable and grassland were evaluated moderately negatively (utility of –14.91). This pattern indicates that siting on brownfields or already impacted land is central to acceptance, whereas intrusions into high-value nature areas are seen as least acceptable.

Biodiversity-related design choices also play a crucial role. Projects including extensive biodiversity measures received clearly positive utilities (28.56), “some measures” were also valued positively (9.01), and scenarios without any biodiversity measures were strongly rejected (–37.57). Visual impacts show a similar gradient: installations visible from living areas were disliked (–32.30), while those only visible from paths and roads received slightly positive evaluations (9.47), and non-visible installations were preferred (22.83), underlining the continued importance of landscape aesthetics in renewable energy siting debates.

Energy justice features are somewhat less influential than siting and biodiversity. A citizen participation model with a share of returns was clearly preferred (13.10), a one-off payment to affected residents was evaluated negatively (–12.37), and membership in an energy community was essentially neutral (–0.72). For procedural justice, pure information-only processes were seen negatively (–13.95), while opportunities for feedback (5.03) and especially formats that grant residents a tangible say in decisions (8.92) increased support. Taken together, these findings highlight that respondents value meaningful participation and ongoing, collective benefit-sharing more than one-time compensation.

The regression analysis confirms that psychological and social factors systematically shape support for USPV in biosphere reserves. The overall model explained a substantial share of variance in support ($R^2 = .571$, adjusted $R^2 = .551$). Intentions to engage in renewable energy activism emerged as the strongest positive predictor of support ($\beta = .418$, $p < .001$), followed by perceived social norms in favour of renewable energy technologies ($\beta = .221$, $p < .001$). Positive perceptions of USPV impacts ($\beta = .114$, $p = .003$) were associated with higher support, whereas negative impact perceptions were associated with lower support ($\beta = –.121$, $p = .004$). Other variables, including general environmental concern, climate concern, conspiracy beliefs, energy justice, and place attachment, were not statistically significant once these core factors were controlled for. This points to the importance of social embeddedness, perceived collective endorsement, and concrete impact appraisals over more general environmental attitudes in shaping support for USPV.

Through **qualitative interviews** we aimed to gain an in-depth understanding of local citizens’ and stakeholders’ experiences, preferences and problem perceptions regarding ground-mounted photovoltaics (GMPV) in biosphere reserves. We further investigated questions of distributive, spatial, procedural and recognition justice in the context of GMPV.

In keeping with the survey results, the interviewees frequently addressed the spatial siting of GMPV and demonstrated a strong preference for installing PV on already built-up or contaminated areas (such as rooftops, parking lots or brownfields) rather than on open land. Many felt that agriculturally valuable land, in particular, should be reserved for agricultural production. However, some interviewees considered GMPV an opportunity for land owners

to turn profit from areas with challenging conditions for agriculture or maintenance, such as steep hillsides or sites with poor soil quality.

In addition to agriculture, participants discussed nature and biodiversity conservation objectives as potential areas of land use conflict or, in some contexts, synergistic effects. Many felt that biodiversity impacts of GMPV should be assessed by experts before making siting decisions but expressed confidence that this dimension is already well-represented in existing spatial planning procedures. In contexts where agricultural landscapes are perceived to be of little ecological value, such as the large corn fields around the biosphere reserve Unteres Murtal, the potential co-benefit of promoting biodiversity through GMPV was especially emphasized by many interviewees. Overall, the majority of interviewees expressed interest in dual land use formats combining GMPV with either agricultural activities (Agri-PV) or biodiversity conservation efforts and argued that synergistic effects could raise public acceptance of GMPV installations.

Further to these competing land use demands, interviewees also frequently addressed the visual impacts of GMPV installations on the landscape, although the opinions on and perceptions of these impacts varied widely among participants. Two context-dependent aspects of GMPV's visual impacts emerged through the interviews: First, impacts depend on landscape conditions, as GMPV installations are much less conspicuous in flat landscapes such as around the biosphere reserve Unteres Murtal than e.g. in the Lungau region, which is characterized by hilly landscapes. And second, the significance of landscape characteristics and existing land use forms for the local residents' regional identity plays an important role for their perception of GMPV. Taken together, these aspects manifest in a preference for smaller GMPV installations that are perceived as less disruptive in small-scale landscapes such as Lungau, and many interviewees articulated a desire for visual integration of GMPV into the landscape (e.g. by following the lines of hills and surrounding installations with hedges or other barriers). In this context, distance of GMPV to existing infrastructure or settlement areas was discussed controversially, as more distant installations might be better hidden from view, but proximity to settlement areas could reduce the need for additional new infrastructure (such as access roads).

While the spatial dimension of GMPV was addressed most frequently in the interviews, the distribution of financial benefits also played an important role. Envy over profits from GMPV was frequently mentioned as relevant for potential opposition to projects, and opportunities for citizens to invest in planned GMPV installations themselves and share in the profit were seen as potential solutions to this as they could contribute to a fair distribution of monetary benefits among those affected by visual, land use and landscape impacts.

On the other hand, citizen participation in procedures of decision-making around GMPV was often discussed critically, as many interviewees expressed high trust in existing governance structures and procedures to represent different interests and saw direct involvement of citizens as a potential complication of the decision-making process. However, local residents in particular also expressed a wish for more transparent decision-making and planning procedures. Central to this is the active communication of information to citizens and stakeholders at an early stage of a project, as early opportunities for participation are key to ensuring planning and investment security both for residents as well as for potential GMPV owners. Regarding the intensity of broad citizen participation in planning processes, the

interviewees expressed differing opinions, with some focusing on increased legitimacy and acceptance through participation, while others felt that the increased complexity of a highly participatory process may outweigh its benefits.

In this context, many interviewees also expressed a preference for locally or regionally owned GMPV installations, since large (inter)national investors were considered to be mainly profit-focused and less considerate of local needs and specific circumstances. In general, the goals and intentions behind GMPV expansion were frequently mentioned as relevant to public opinion: Many interviewees rejected pure profit orientation but showed support for goals such as energy independence or self-sufficiency. However, this support was often restricted to the regional level and a contribution to national targets beyond the regional demand was discussed more controversially, with many interviewees suggesting that it would be better to produce electricity closer to where it is used or needed.

Discussion

The findings are consistent with other work showing that land-use conflicts and biodiversity concerns are central to public debates about ground-mounted PV, particularly in ecologically sensitive or protected landscapes (Biró-Varga & Stremke, 2026; Hilker et al., 2024; Roddis et al., 2020; Rösch & Fakharizadehshirazi, 2025, 2025). Scholarly work from Central Europe has emphasised that site selection is of utmost importance when aiming for a balanced approach between biodiversity conservation and the installation of additional utility scale photovoltaic capacities (cf. Hainz-Renetzeder et al., 2025). which is mirrored by the strong preference for sealed and pre-impacted sites observed here. Just like a recent study from Germany has confirmed (Rösch & Fakharizadehshirazi, 2025) method to reduce ecological impacts, such as siting on brownfields, alongside transport corridors, or already degraded land can also improve public acceptance, even though the ecological effectiveness of many measures still requires robust long-term evidence (Hainz-Renetzeder et al., 2025; OECD, 2024). The pronounced rejection of projects on ecologically valuable land however aligns with recent calls to apply the precautionary principle and to “mainstream biodiversity” into renewable power planning and to systematically avoid high-biodiversity habitats when expanding solar infrastructure (Hainz-Renetzeder et al., 2025; OECD, 2024)

International policy-oriented assessments highlight the potential of co-location approaches (such as agrivoltaics and multi-functional land use) to reconcile climate and biodiversity goals, and the present results suggest that citizens are responsive to such attempts when they are made visible and credible in project design. The use of virtual reality renderings in this study further connects to a growing literature on visualisation tools (e.g., PPGIS, scenario maps) that enhance laypeople’s understanding of land-use implications and can support more informed and equitable local planning.

In terms of social acceptance and energy justice, the results echo international research showing that meaningful participation, transparent procedures, and fair benefit-sharing arrangements increase perceived legitimacy and support for renewable energy projects. The preference for citizen participation models with a share of returns over one-off payments is in line with evidence that long-term, collective benefit schemes are perceived as more just and

less “buying off” opposition. The high importance of social norms and activism intentions as predictors of support also dovetails with behavioural and social-psychological studies that emphasise the role of normative climates and collective engagement in energy transitions, suggesting that communication strategies and participatory formats that foster pro-renewable norms and empower local actors can be particularly effective. Overall, the study contributes to international debates by linking ecological siting criteria, biodiversity measures, and energy justice dimensions in a single empirical design focused on biosphere reserves, a context where the tensions between renewable expansion and conservation are especially pronounced.

Publications

Based on the work in this package, the following scientific outputs are envisaged or already underway:

- Conference Vilnius
- Conference Graz
- Sposato, R., Klic, R., & Hampl, N. (submitted / in preparation). “Green Energy, Greener Habitats? Public perceptions of utility-scale photovoltaics in biodiversity landscapes.” Target journal: peer-reviewed energy or environmental social science journal.
- Master thesis Michaela Niederhofer
- planned publication based on interviews

Data and methods

Data were collected via a structured online survey with an embedded CBC experiment administered in January 2025 to a quota-based sample of Austrian residents living in or near three biosphere reserves. Recruitment and incentives were handled by a professional market research panel provider, and all participants gave informed consent prior to participation. Quality controls included an RLH-based exclusion of near-random conjoint responses (threshold 0.34) and removal of extremely fast completions (below 50% of median completion time), resulting in a final analytical sample of 512 respondents and a median survey duration of about 13 minutes. This procedure ensured a high-quality dataset suitable for both conjoint and regression analyses.

The questionnaire combined socio-demographic items (age, gender, education, political ideology, postal code) with psychometric scales capturing environmental concerns, familiarity and place attachment to the nearest biosphere reserve, support for different renewable energy technologies, perceived positive and negative impacts of USPV, activism intentions, energy justice orientations, and beliefs and norms around renewable technologies. Exploratory factor analyses were conducted to structure multi-item scales, yielding reliable subscales (Cronbach’s alpha generally ≥ .69) for environmental concerns, perceived impacts, beliefs and norms, and energy justice dimensions. The CBC experiment varied five attributes of hypothetical USPV projects—biodiversity measures, visibility, land use type, distributive justice mechanism, and procedural participation format—each with

three levels, across 12 choice tasks per respondent, supported by short textual explanations and VR-based visualisations of key attributes.

Conjoint data were analysed using standard hierarchical or mixed logit-based preference estimation procedures that provide part-worth utilities and relative importance scores for each attribute, allowing a fine-grained assessment of trade-offs in public preferences. To explain support for USPV in biosphere reserves, multiple linear regression models were estimated with support as the dependent variable and psychological, social, and justice-related variables as predictors, including checks for multicollinearity and residual autocorrelation. Together, these methods provide a robust, theory-informed and empirically grounded basis for deriving recommendations on where and how to design USPV projects in biodiversity-sensitive regions such as biosphere reserves.

To investigate individual experiences, perceptions and local knowledge around ground-mounted photovoltaic (GMPV) installations in Austrian biosphere reserves, 8-9 qualitative, semi-structured expert interviews (Meuser and Nagel, 2009) per study region were conducted in July 2024. An interview guide was developed based on a review of the relevant literature, in particular focusing on factors influencing the social acceptance of renewable energy infrastructure as well as the literature on energy justice.

The interviews were conducted as walking interviews, a method which is particularly suited to exploring topics and issues with a strong spatial component, as it facilitates participants' interaction with and access to spatial dimensions of their experiences and preferences (Muhr et al., 2024). The interview is conducted while walking together along a route chosen by the interviewee, which encourages self-reflection and can foster collaborative knowledge production by building trust between the researcher and interviewee and shifting power relations between them (Muhr et al., 2024).

To narrow down the study areas, we chose 3-4 municipalities per biosphere park based on WP2's first tentative assessment of PV potentials as well as consultation with our contacts in the biosphere park management. To create a list of potential participants, we requested suggestions for relevant interview partners from biosphere park managers as well as the selected municipalities' administrations. In addition, we contacted regional and local organizations such as nature conservation, tourism, hunting and farmers' associations, youth groups and energy communities. To cover different perspectives, we aimed to conduct half of the interviews in each study area with stakeholders or decision makers and the other half with citizens. The final sample of participants covered a wide range of backgrounds and interests, although it must be noted that women and younger participants (<30) were much harder to recruit and are thus underrepresented in our sample. Table x shows details of the interviews conducted in the three study areas.

Table x: Overview of the walking interviews conducted in the three case study regions.

	Lungau	Unteres Murtal	Wienerwald
Number of interviews	8 (4+4)	9 (5+4)	8 (4+4)

Time frame	22.-25.7.2024	23.-26.7.2024	11.-30.7.2024
Local residents' backgrounds	agriculture | social profession | local business (electrical installations | recreation) | nature conservation	agriculture | military | beekeeping | forestry | healthcare profession | nature conservation | biology | banking | energy community	retirement | beekeeping | hunting | agriculture | journalism | village renewal | small trade | energy community
Local stakeholders' backgrounds	community politics | tourism | regional management | biosphere reserve management	community politics | agriculture | hunting	community politics | regional management | energy sector | agriculture | hunting

The interviews were recorded and subsequently transcribed using the open-source AI-tool "noScribe"[2]. The transcripts were checked for accuracy and corrected by the researchers where necessary and were then imported into MaxQDA for coding. We conducted a qualitative content analysis (Mayring and Fenzl, 2019) along key dimensions of energy justice, combining deductive coding based on the energy justice literature with inductive coding where appropriate.

References:

Mayring, P., Fenzl, T., 2019. Qualitative Inhaltsanalyse, in: Baur, N., Blasius, J. (Eds.), Handbuch Methoden der empirischen Sozialforschung. Springer Fachmedien, Wiesbaden, pp. 633–648. https://doi.org/10.1007/978-3-658-21308-4_42

Meuser, M., Nagel, U., 2009. Das Experteninterview — konzeptionelle Grundlagen und methodische Anlage, in: Pickel, S., Pickel, G., Lauth, H.-J., Jahn, D. (Eds.), Methoden der vergleichenden Politik- und Sozialwissenschaft: Neue Entwicklungen und Anwendungen. VS Verlag für Sozialwissenschaften, Wiesbaden, pp. 465–479. https://doi.org/10.1007/978-3-531-91826-6_23

Muhr, M., Gartner, F., Scherhaufer, P., 2024. Walking interviews: Exploring urban heat on the go, in: Kogler, R., Hamedinger, A. (Eds.), Interdisziplinäre Stadtforschung II. transcript Verlag, pp. 53–66. https://doi.org/10.1515/9783839471562-003

[2] https://noscribe.de/de/

4.4. WP5 - BioPV-Laboratories

Main Results and findings

Ground-mounted PV is a controversial topic when it comes to the realisation of projects at local and regional levels. Main topics are conflicts with agricultural production and biodiversity, as well as expected negative impacts on landscapes and therefore on recreational values and tourism. Participation is seen as a key factor in the process of finding sustainable solutions with increased acceptance (Campos et al. 2025; Scherhaufer et al. 2017). With the BioPV-Laboratories a novel approach was developed and tested for collaborative PV scenario development in three different biosphere parks. Results show that the involvement of a broad range of people with different backgrounds requires comprehensive datasets and low-barrier tools for interaction. The laboratories emphasize the strong interconnection among ecological, economic, technical, political, and social aspects.

Material and Methods

A data-driven simulation game approach was used to facilitate computer interaction in diverse groups and to allow (1) the collaborative development of Go- and No-Go-areas for ground-mounted PV, (2) the virtual establishment of ground-mounted PV installations with different settings (size, density, ground management, fencing) and (3) the assessment of planning decisions based in indicators and immersive 3D visualisations.
Data from the workshops are logged using written protocols and are linked with the spatial data (Go-/NoGo-areas and PV scenarios) for further analysis and visualisations (see results)

Technological Framework: Web-GIS and Landscape Lab

By developing a technological framework for joint action, we have created several opportunities to collect data and contributions from a region. For low-threshold data collection, a prototype for a WebGIS (see Fig. XXX)was developed to collect zones for or against the expansion of ground-mounted PV at a low threshold at local or regional level. The advantage is that many people can be reached by using suitable communication channels. A disadvantage could be that the information content of this type of query is often low and requires greater effort for quality assurance (filtering out incorrect or meaningless data points).

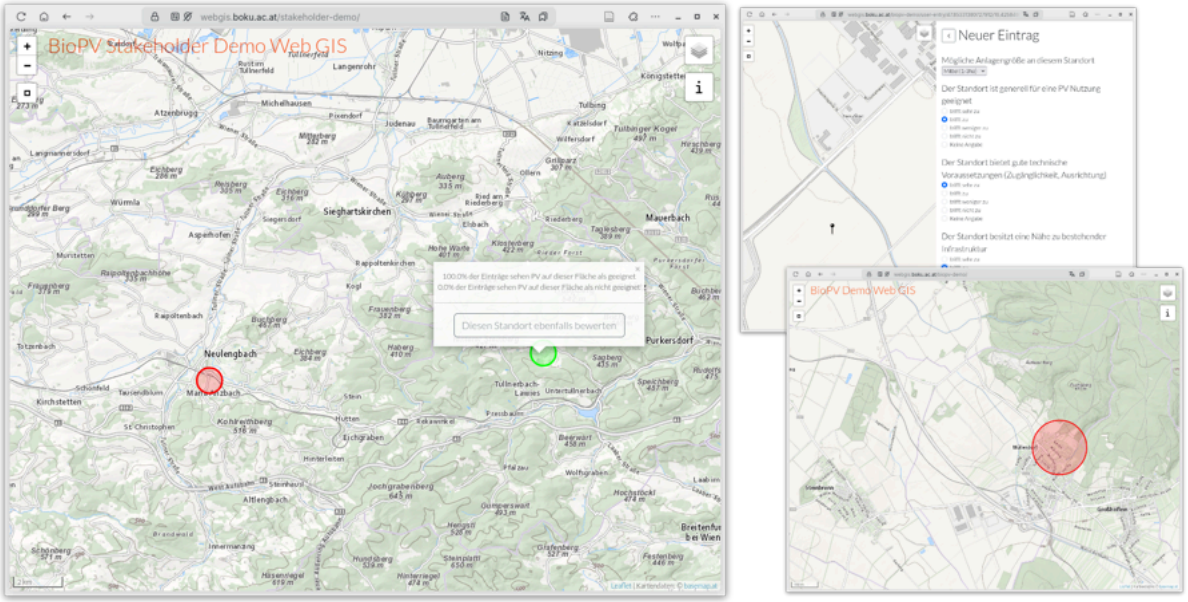

Fig. XXX: Prototype of the WebGIS application for collecting zones for or against the expansion of ground-mounted PV

Since the qualitative reasons for or against ground-mounted PV at a specific location were of great importance for the project, we focused on data from the on-site workshops, as the arguments presented there were more in-depth and detailed.

To make planning accessible and allow a wide range of people to interact with the trade-offs of PV planning, a custom toolkit for immersive landscape visualization and serious game development, the *LandscapeLab*, is employed. This toolkit aims to allow both experts and laypeople to plan virtual PV installations and get immediate feedback on the economic cost and value, the ecological impact, and the visual landscape impact. The findings and datasets from WP2 and WP3 serve as the basis for these impacts; the *LandscapeLab* communicates them to participants and allows them to make specific decisions based on their personal evaluation of the various impacts and their regional knowledge.

The *LandscapeLab* allows workshop participants to interact with a virtual map (projected onto a table) using simple game bricks (see Figure XXX, left). One brick may represent a Go- or NoGo-area as well as a PV installation to be placed at that location. Furthermore the bricks can be used to navigate on the map and determine the viewing location for the 3D environment. Additional parameters can be configured in a popup menu for a planned PV plant (see Figure XXX, left), and indicators can be shown to provide additional information based on its location and configuration. Created PV plants are immediately visualized in an immersive 3D landscape visualization, which participants can view on a projector or via virtual reality (VR) glasses (see Figure XXX, right).

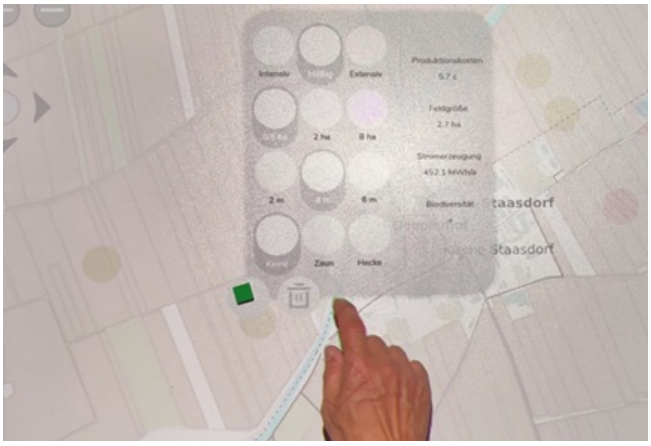

Figure XX: virtual map, game brick, and parameter
3D visualization on a screen and immersive VR experience (right).

The toolkit is developed with the game engine *Godot*, which provides a basis for high-quality 3D visualization with full interactivity and support for VR devices. A custom plugin is used to directly load GIS data (open government data and data from WP2 and 3) into the engine for visualization and interaction. The most important data sources are:

- Digital Terrain Model and Digital Elevation Model (raster, 1m resolution, source: BEV)
- Land Cover (raster, 1m resolution, source: INVEKOS, Copernicus, OpenStreetMap)
- Building polygons (vector, source: OpenStreetMap)
- Road network (vector, source: OpenStreetMap)

The LandscapeLab, based on previous works (Schauppenlehner et al. 2019, Schauppenlehner et al. 2020), has made significant advancements for allowing nuanced and parametric visualization of PV installations and to incorporate the complex game logic. These developments were embedded in the inter- and transdisciplinary research process: project members of other workpackages were regularly asked for feedback and involved in testing, and a prototype of the toolkit was demonstrated in the second stakeholder workshop (27.02.2024), where the stakeholder group was able to play the serious planning game and provided valuable feedback on the tool. The resulting technology and configuration for visualization and interaction is described in section XXXX

Case-study laboratories

Five moderated workshops were conducted in the biosphere park regions of Lungau, Unteres Murtal, and Wienerwald using the technological framework described above, to examine techno-economic, ecological, and societal aspects of site selection and the expansion of ground-mounted photovoltaic systems at the regional level. In total, 39 individuals participated, including citizens and stakeholders from various fields such as regional management, nature conservation, renewable energy, agriculture, spatial planning, and local politics. A stakeholder analysis was carried out for each location, resulting in a list of potential stakeholders that served as the basis for sending out invitations. Additionally, key stakeholders were encouraged to recommend other relevant participants and help promote the workshops. Case-study laboratories aimed at citizens were advertised through the

websites of the biosphere parks and local communities where the events were held, as well as through local stakeholders. Each case-study laboratory included six to ten participants to ensure an interactive and collaborative planning process using the immersive 3D simulation game.

The simulation game offers the opportunity to interactively explore various locations for ground-mounted PV, as well as different expansion options, system sizes, specification options, and management strategies. In the first phase of the game, potential suitable and exclusion zones for the expansion of ground-mounted photovoltaic systems were identified, taking economic, ecological, and societal aspects into account. Subsequently, different expansion scenarios and design options for the systems were visualized interactively. Participants were able to evaluate the selected options and discuss the impacts of these scenarios in relation to the goals of the energy transition, biodiversity effects, and societal considerations. Additionally, questionnaires were used to gather participants' attitudes, knowledge, and engagement regarding renewable energy, as well as their feedback on the workshop.

Workshop stages

- Welcome and Introduction: brief introduction of the participant and overview of the objectives of the project, workshop goals and the procedure
- Baseline survey: an initial questionnaire captured participants' engagement with renewable energy and attitudes toward ground-mounted PV, along with demographics.
- Immersive 3D simulation game: two-phase process using PPGIS and VR enabling participants to engage in interactive scenario planning
- Reflection: recap and discussion of the key outcomes and insights
- Debrief and feedback questionnaire on the experience and learning outcomes

Figure XX: Impressions from the case-study laboratories: Collaborative work on the interactive game board table (left), and immersive 3D experience using VR-glasses (right)

4.4.1. Results

The results show that suitable tools and methods can promote cooperation between different participant groups especially in the field of renewable energy planning. Technological advances in high-resolution geodata and advanced hardware are opening up new opportunities for cooperation with the support of data-based approaches.

4.4.1.1. Technological Framework and Evaluation

Using the LandscapeLab as the basis, a comprehensive PV simulation game was developed. The game combines the outputs from other workpackages and allows players to interact with them by siting and configuring photovoltaic plants, based on a game logic developed in an iterative inter- and transdisciplinary process together with project partners and stakeholders. This method and result is described in detail in an upcoming publication in the 2026 issue of the Journal of Digital Landscape Architecture (Bittner, Baumgartinger-Seiringer & Schauppenlehner, 2026). A brief overview is given here in figure XXX.

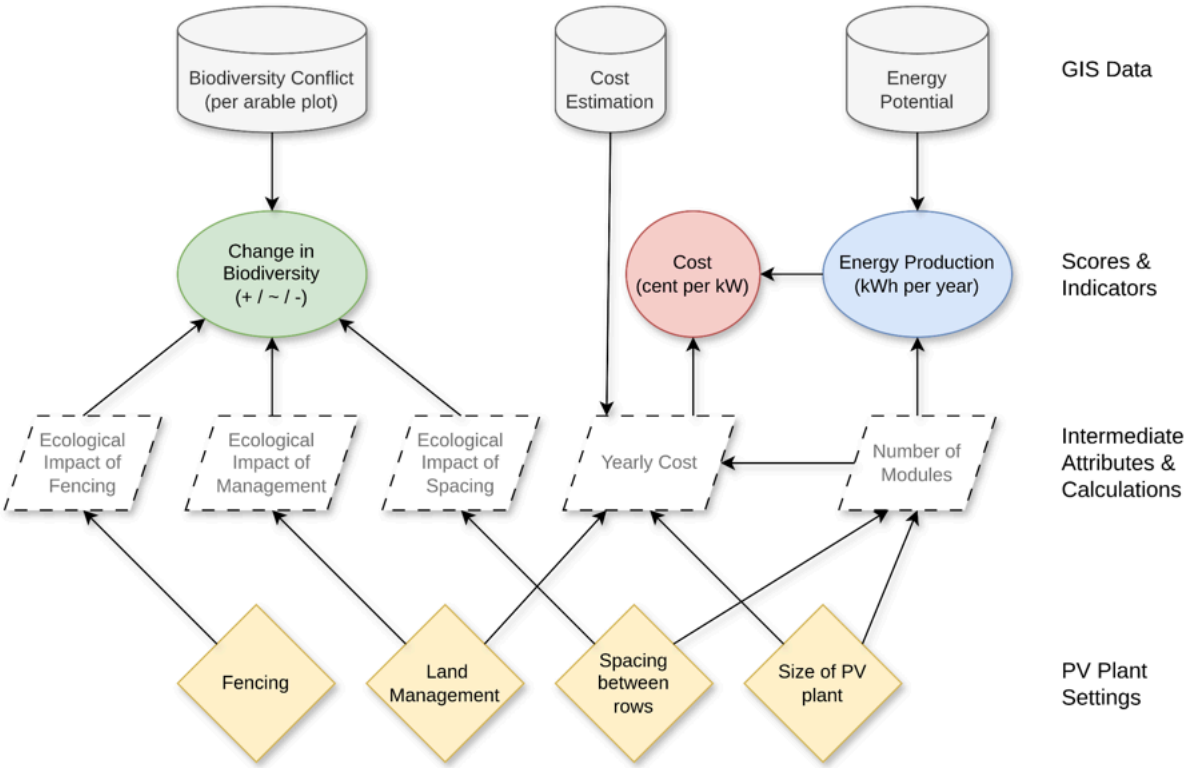

Figure XXX: Flow chart diagram of the serious game logic, showing how GIS datasets are connected to attributes and user inputs.

Ecological data was integrated by assigning an ecological potential for conflict (EPFC) to each INVEKOS plot as part of the spatial analysis in WP3. Each configurable PV option was then assigned a biodiversity impact (positive, negative, or neutral) depending on the EPFC of the previous land-use. For example, extensive ecological management causes a positive biodiversity impact on fields with low or medium EPFC, but on fields with a high EPFC, the impact is still negative.

Using the output from WP2, the total energy production is calculated for each fictional PV plant designed in the simulation game. By dividing the energy production by the total cost of

this PV plant (also based on data from WP2), the energy price in ct/kWh is calculated and communicated to the participants on the interactive gameboard.

To accurately communicate the visual impact of a PV plant on the landscape, each configurable option is reflected accurately in the landscape visualization. This required particular care with the ecological management options: based on the field surveys from WP3 and iterative testing with the project partners, plants and ground textures were defined for each different management option, differentiating between areas underneath, in front of, behind, and around PV modules (see Figure XXX).

Figure XXX: Visualization of different management methods of PV fields: intensive (left, graveled and regularly mowed, distance of 2m between modules), medium (center, mowed and mulched, 4m between modules), and extensive (right, measures taken to promote flowering plants, 6m between modules).

The technological framework and graphical implementation of the *LandscapeLab*, used as a planning tool during the case-study workshops, were rated very positively by participants. Over 90% found the graphical quality of the 3D visualization appealing and realistic. Participants considered the interactive map easy to use and were able to navigate and orient themselves well within the 3D environment. Overall, the planning game was perceived as engaging and inspiring, contributing to participants' understanding of the potential and impacts of ground-mounted PV (Figure XX). The results of the case-study workshops demonstrated that data-driven methods and extensive visualization capabilities enable active exploration of the impacts of ground-mounted PV installations. Moreover, the gamified approach lowers entry barriers and promotes peer-level dialogue between laypersons and experts.

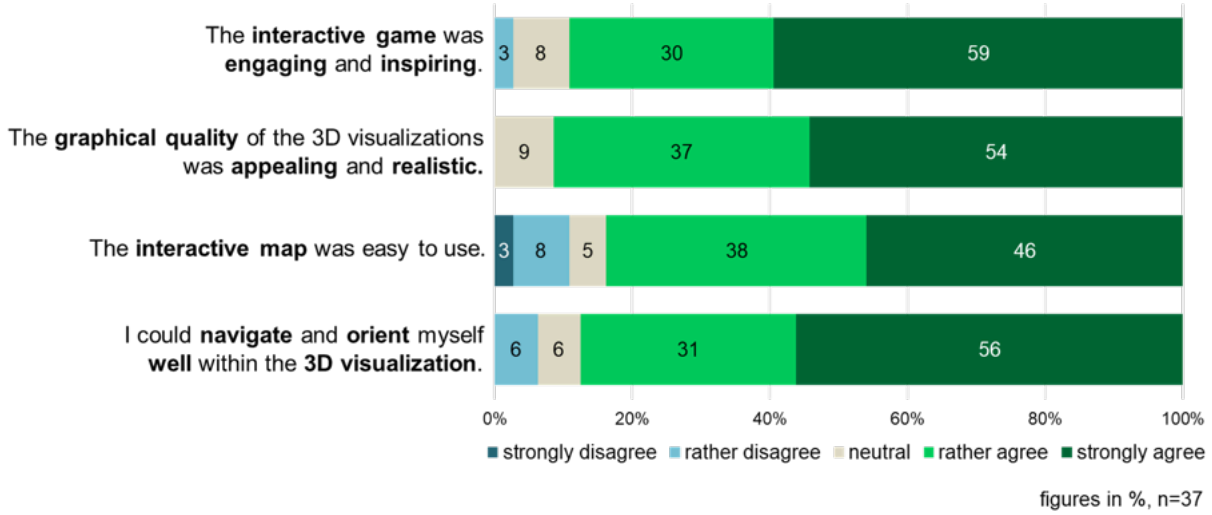

Figure XX: Workshop participants' feedback on the planning game, visualization, and user experience of the *LandscapeLab*.

4.4.1.2. Results of the case-study laboratories

The results of the case-study laboratories emphasize the strong interconnection among ecological, economic, technical, political, and social aspects, and show how different, sometimes conflicting, interests intersect (Figure XX). The key criteria and decision making factors can be grouped into three main areas: **landscape and ecology, society, and technology and economy**.

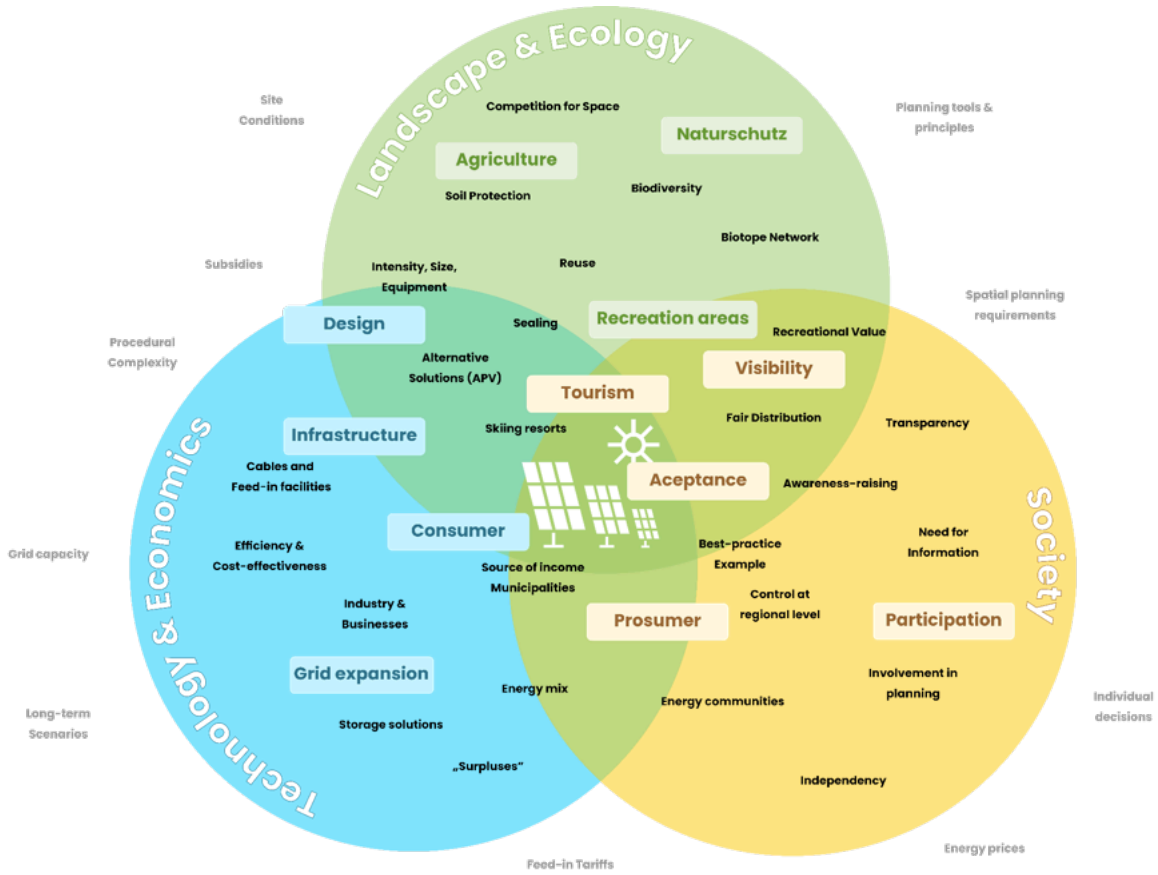

Figure XX: Key areas of decision making and site selection criteria for the expansion of ground-mounted PV resulting from the case-study laboratories.

Prominent topics discussed during the laboratories included the competition between renewable energy production, particularly ground-mounted photovoltaic expansion, and agriculture, with an emphasis on protecting high-quality soils. Other important issues included the visibility of ground-mounted photovoltaic installations in and around residential areas, recreational spaces, and tourist destinations, as well as the impacts on biodiversity. Participants also recognized opportunities to use PV sites to improve habitat connectivity. Proximity to existing energy infrastructure and its implications for grid expansion and associated costs were identified as critical factors in all three project areas. There was also consensus that suitable zones for new installations should be located near existing infrastructure such as ski lifts and transport infrastructure, for example along motorway and railway corridors. By comparison, perceptions of the landscape's aesthetic and ecological value varied regionally. In regions with higher tourism and recreational use (such as Lungau and the Wienerwald), the landscape aesthetics and visibility of PV installations were discussed frequently, whereas these aspects played a subordinate role in regions characterized by intensive agriculture and the presence of several large ground-mounted PV installations, such as the Murtal region.

Additionally, participants highlighted the importance of integrating renewable energy into spatial planning, as this provides an essential basis for greater planning certainty and more informed decision-making. In regions with more defined zoning frameworks (such as the Unteres Murtal region), the designated zones were a central focus of discussion, and plans

to expand ground-mounted PV relied heavily on those designations. Within the designated zones, however, development was pursued relatively liberally, with participants tending to plan larger and more numerous sites. By contrast, in regions with fewer prescriptive guidelines, discussions about suitable areas were more spatially open, yet planning proceeded much more cautiously.

With regard to the planning of specific facilities, the arguments put forward were largely taken into account. Below is an example from the Lower Murtal Biosphere Park (figure XX). Two PV plants (8 ha and 21ha with a 4m distance between the rows) were planned on a prioritized area according to the “Sachprogramm Solarenergie”. By selecting an intensively used agricultural area, extensive management strategies and no fencing, an increase in biodiversity measures can be expected. Due to the nearby forest, field tree structures, and the even topography, the facilities can only be seen from a few points.

Figure XX: Virtually planned PV plant in the Lower Murtal Biosphere Park

The findings underscore that public acceptance can be significantly strengthened by creating opportunities for participation in the planning process and by ensuring that the energy generated is used locally. Participants emphasized the importance of co-creation and transparency in planning processes, as well as fair allocation between municipalities and

regions, strong acceptance of self-consumption and energy communities and discussed regional self-sufficiency, and questions of social justice (where, by whom, and for whom).

The learning experience in the case-study laboratories was rated very positively by participants. The workshops enhanced understanding of the impacts of ground-mounted PV and fostered collaborative, peer-to-peer learning. Nearly half of participants reported that the workshops influenced their attitudes toward ground-mounted PV. In addition, 86% advocated for more initiatives of this kind in their region (Figure XX). Further potential application areas for the tool included wind energy, habitat connectivity, spatial planning, tourism infrastructure, transport infrastructure, healthcare services, and housing projects and their impacts.

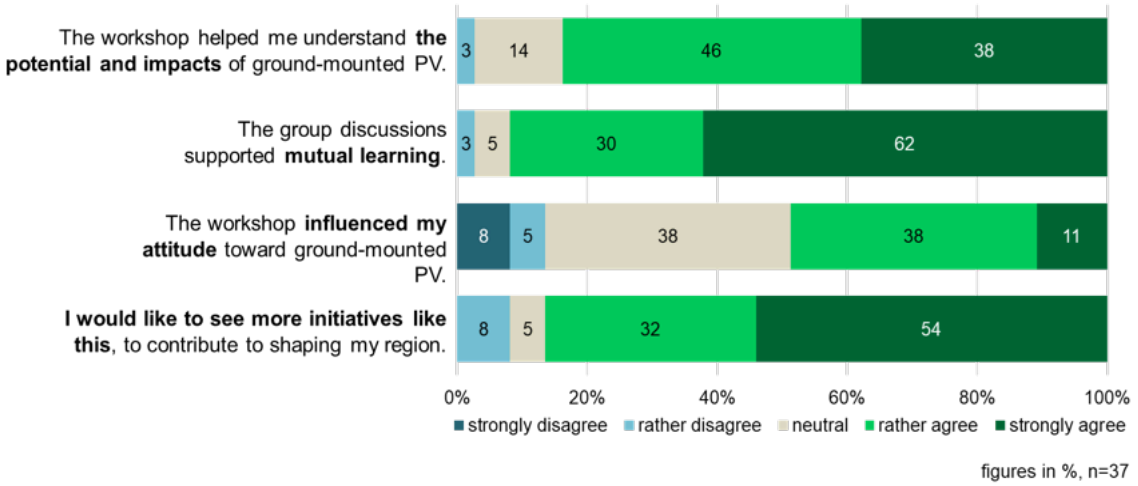

Figure XX: Participant's feedback on the learning experience from the case-study laboratories.

- Publications – list of all published or submitted paper, date, journals, DOI

4.5. Interdisciplinary Integration

4.6. Comments from Mid-term Evaluation

4.7. Policy Recommendations (if applicable) and potential impact

5. Added Value of transdisciplinary contributions

- Stakeholder Engagement – Involvement, contribution, and impact
- Knowledge Co-Creation
- Societal Impact - Sustainability and Transformation Pathway

6. Public communication and outreach

- Non-Technical Publications – non-peer-reviewed publications and website

Hainz-Renetzeder, C; Karner, K; Milchram, M; Petermann, J; Strohmaier, B; Tribsch, A; Weber, N. (2025). Fact Sheet #03: "Biodiversitätsverträglicher Ausbau von PV-Freiflächenanlagen" [pdf]; https://www.biodiversityaustria.at/wp-content/uploads/2025/04/BD-Hub_Oe_Factsheet_03_Biodivfreundlicher_PV-Ausbau.pdf

Hainz-Renetzeder C, Pospisil K, Pachinger B (2024). Grüne Energie, reiche Insektenwelt? Biodiversität in Freiflächen-Photovoltaik-Anlagen. Zoll+ Landschaftsplanung und Landschaftsökologie 44:56-60.

- Teaching

“Ground-mounted PV-Systems” as part of the interdisciplinary project course for 36 bachelor's students (Bachelor programme Environment and Bioresources Management at BOKU University) in the winter term 2023/24 & 2025/26

“Climate protection - Nature conservation; Expansion of open-space photovoltaics in conjunction with biodiversity protection thinking”; GIS-supported project (Bachelor programme Landscape Architecture and Landscape Planning) Summer term 2024

- Outreach Activities – public lectures, media, exhibitions, etc.

Hainz-Renetzeder, Christa, Pospisil, Katharina, Neumayer, Hans, Pachinger, Bärbel (2025): Ergebnisse der Freilanderhebung und Empfehlungen zur Förderung der Biodiversität, BioPV. Factsheet #2, Universität für Bodenkultur Wien, https://doi.org/10.5281/zenodo.17881407

Hainz-Renetzeder, C.; Pachinger, B. (2025). Grüne Energie: Licht und Schatten für die Biodiversität. Licht und Schatten - Freiflächen-Photovoltaik im Spannungsfeld zwischen Landschaft, Mensch und Biodiversität, Veranstaltung: BOKU Energiecluster 21.10.2025

Hainz-Renetzeder, C; Pachinger, B. (2025). FF-PVA & Biodiversität? BioPV Webinar: Vorstellung der Projektergebnisse 2025. Datum: 09.12.2025

Hainz-Renetzeder, C. (2024). Klimakrise! Biodiversitätskrise! Die naturverträgliche Energiewende als Krisenintervention. Veranstaltung: Nachhaltige Energiezukunft. Landnutzungsänderungen im Fokus der Energiewende – globale und lokale Auswirkungen. BOKU Energiecluster 16.04.2024

- Conferences and Workshops

Hainz-Renetzeder, C; Baumgartinger, M; Bittner, K; Dworzak, V; Hampl, N; Mikovits, C; Pachinger, B; Plöchl, J; Schauppenlehner, T; Sposato, R; Scherhaufer, P. (2024):

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Pachinger, B; Pospisil, K; Neumayer, J; Kropf, M; Scherhaufer, P; Hainz-Renetzeder, C (2024) Greenwashing oder Artenförderung? Insekten in PV-Freiflächenanlage. 90. Internationale Entomologentagung 2024, 08.11.2024 - 10.11.2024 Entomologische Arbeitsgemeinschaft am Biologiezentrum Linz, Linz, Österreich

Pachinger, B; Baumgartinger, M; Bittner, K; Hainz-Renetzeder C; Klic, R; Mikovits, C; Plöchl, J; Schauppenlehner, T; Sposato, R; Scherhaufer, P (2025) Herausforderungen einer integrierten nachhaltigen Entwicklung bei PV-Freiflächenanlagen (BioPV). Vortrag bei den Tagen der Biodiversität 2025 – Gemeinsam aus der Biodiversitätskrise 25.02.2025 - 28.02.2025, Wien BOKU University, Zoologisch-Botanische Gesellschaft Österreich (ZOOBOT), Wien.

- Public Access to outcomes
- Public impact – promotion of public understanding of research, raising of awareness

7. Challenges / Lessons learned

- Challenges encountered

- Lessons for inter- and transdisciplinary research

- Potential integration in future projects

8. Published and submitted papers / other output

Hainz-Renetzeder, C., Schauppenlehner, T., Scherhaufer, P., & Pachinger, B. (2025). Navigating the Nexus between Renewable Energy and Biodiversity: Impacts and Mitigation Strategies for Ground-Mounted Photovoltaic Systems. Landscape Online, 1141–1141. https://doi.org/10.3097/LO.2025.1141
Schauppenlehner, T., Bittner, K., Baumgartinger-Seiringer, M. (2025). Wie sieht die Energiewende jetzt eigentlich aus? KinderUni BOKU 2025

Bittner, K. (2025). Plants, Polygons and Pixels: Large-Scale Vegetation Rendering in Godot. GodotFest 2026. https://www.youtube.com/watch?v=U2nBOOR_PP4

Bittner, K., Baumgartinger-Seiringer, M. Schauppenlehner, T. (2026). Playful Complexity: Using a Serious Game with Immersive 3D Visualization To Balance Economy, Ecology, and Landscape Impact of Photovoltaic Plants. JoDLA – Journal of Digital Landscape Architecture, 11-2026 (in print)